

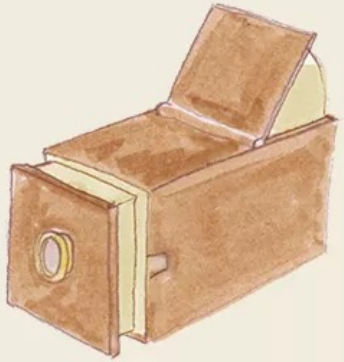
上海科技大学
ShanghaiTech University

Camera

CS172 Computer Vision I

Instructor: Jiayuan Gu

History of Photography from the spruce



Camera Obscura is invented

1021

1685

First portable camera



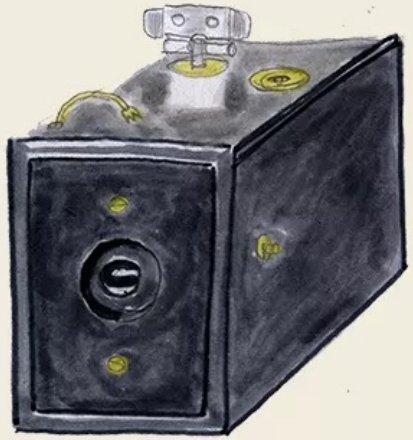


World's first photograph

1826

1888

Kodak sells first commercial camera





WWII helps shape a new style of photography

1939

1948

Polaroid introduces instant image development





First professional digital camera

1991

2000

First camera phone



Figure from the spruce

Agenda

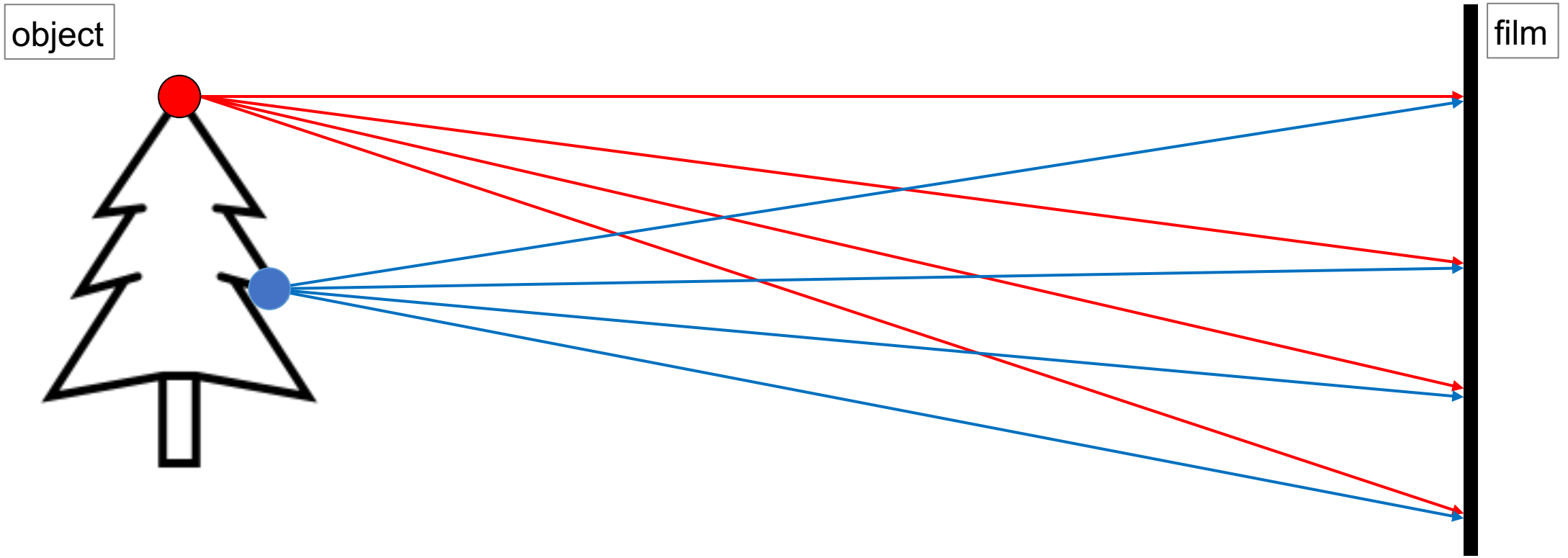
- Learning Objectives
  - Explain why a **pinhole** forms an image
  - Derive and apply **perspective projection**
  - Predict the effects of **depth**, **focal length**, and **aperture**
  - Explain **focus**, **depth of field**, and common **lens deviations**

- Roadmap
  - Ideal geometric camera
  - Practical camera with a lens
  - Where the ideal model fails

Pinhole Camera

Part I: How can a 3D scene form a sharp 2D image?

Image Formation without a Pinhole

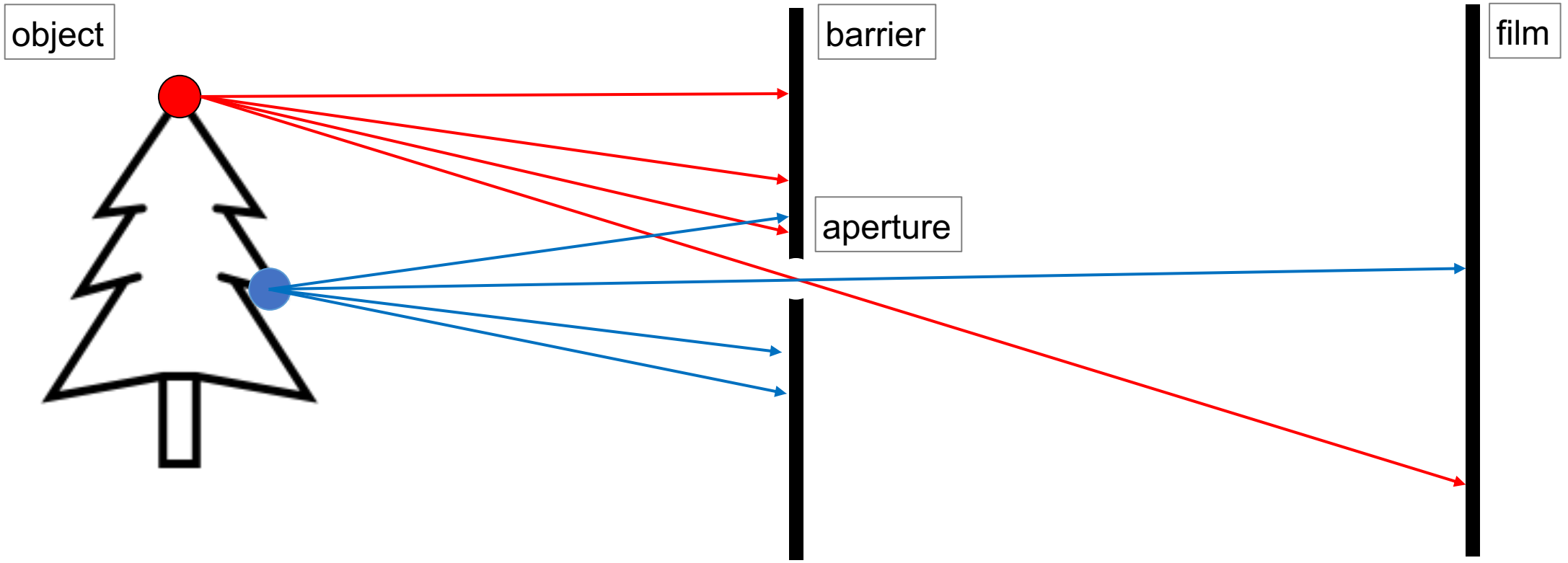


Rays from many scene points arrive at the same sensor location, so spatial correspondence is lost

Without restricting rays, every sensor point mixes light from many scene points

Modified from Steve Seitz

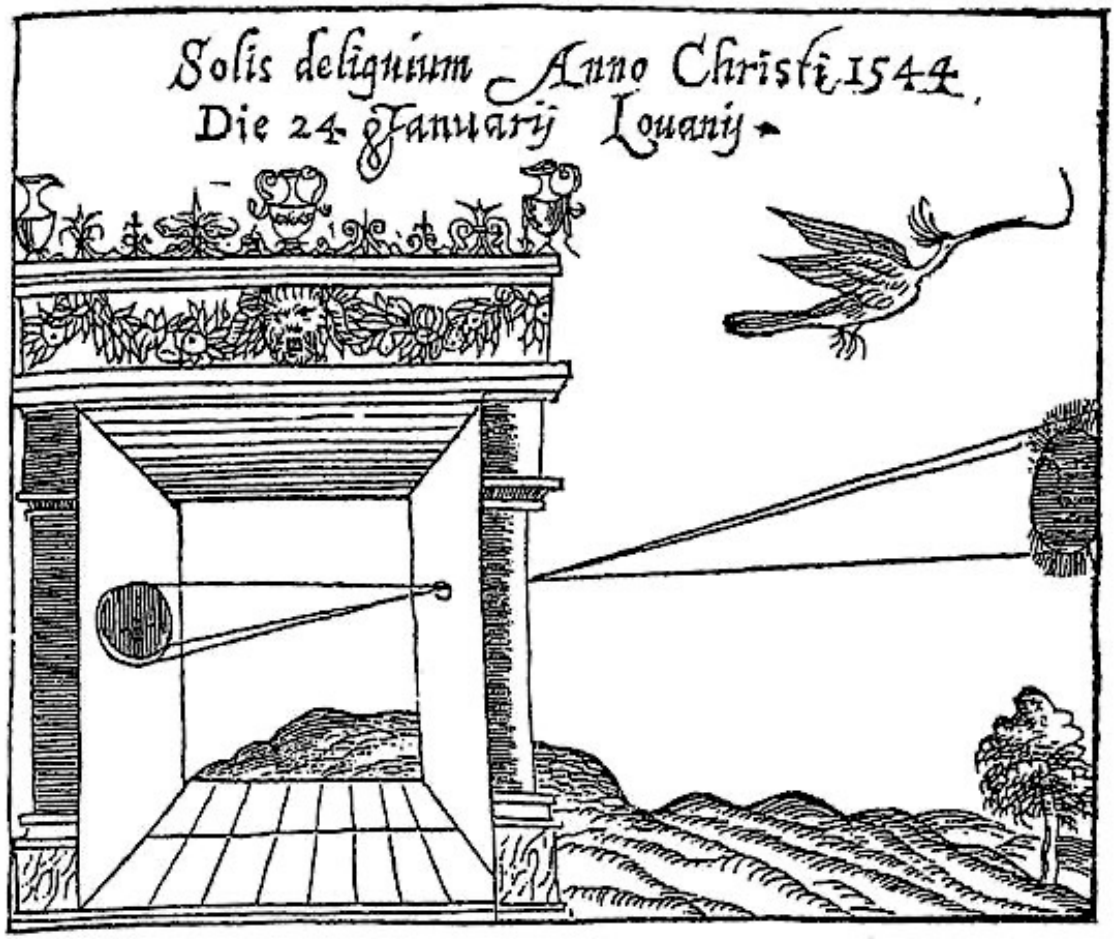
Pinhole Camera



The opening is known as the **aperture**

The pinhole creates a one-to-one correspondence between scene directions and image locations

Camera Obscura



Gemma Frisius, 1558

Brief History

- Mozi (468-376 BCE): early description of inverted image formation
- Aristotle (384-322 BCE): observed small gaps projected images of the crescent sun during a partial eclipse
- Al Hacen (965-1039): wrote *Book of Optics*
- Leonardo da Vinci (1452–1519) recorded a detailed description of the camera obscura in 1502

Modified from A. Efros

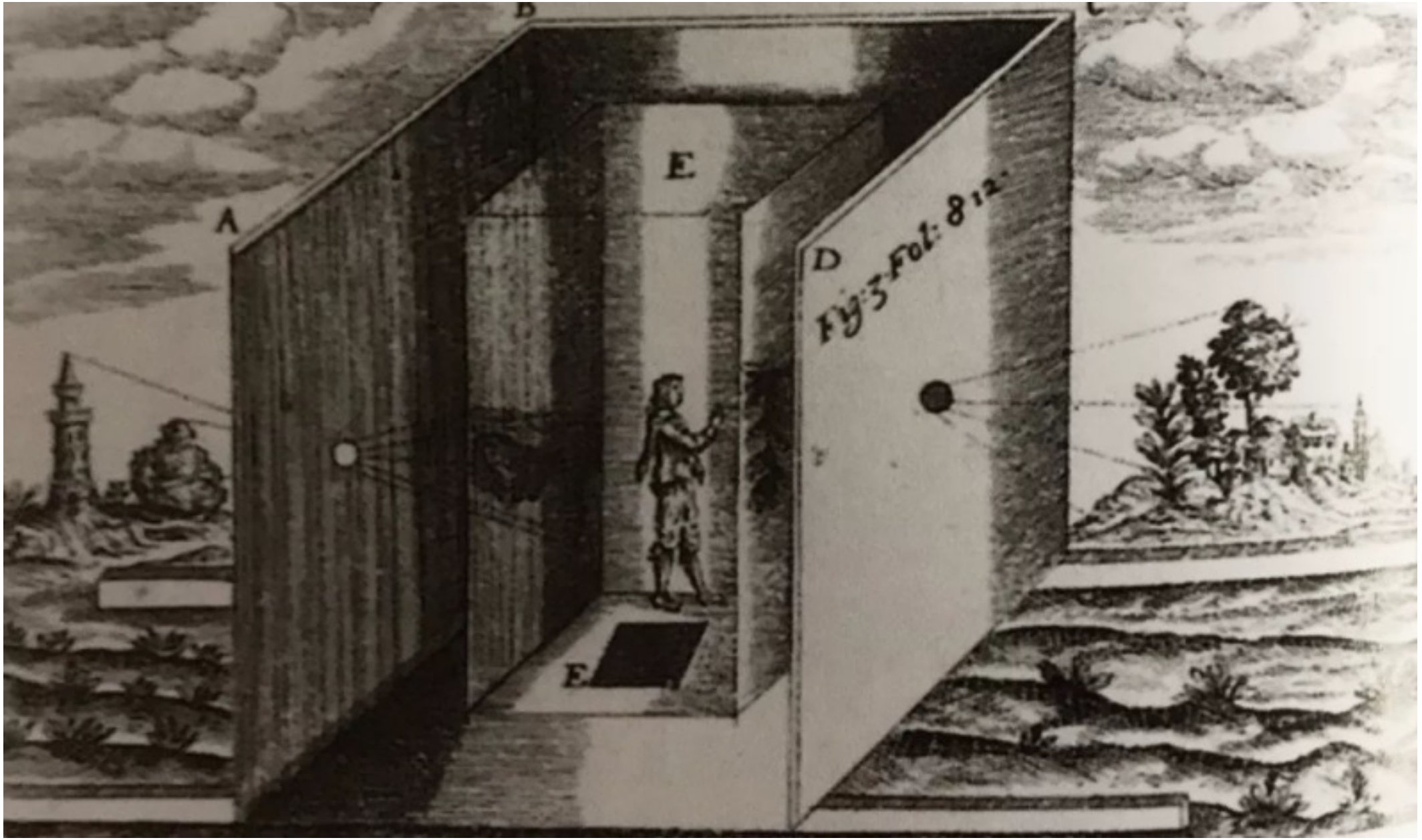
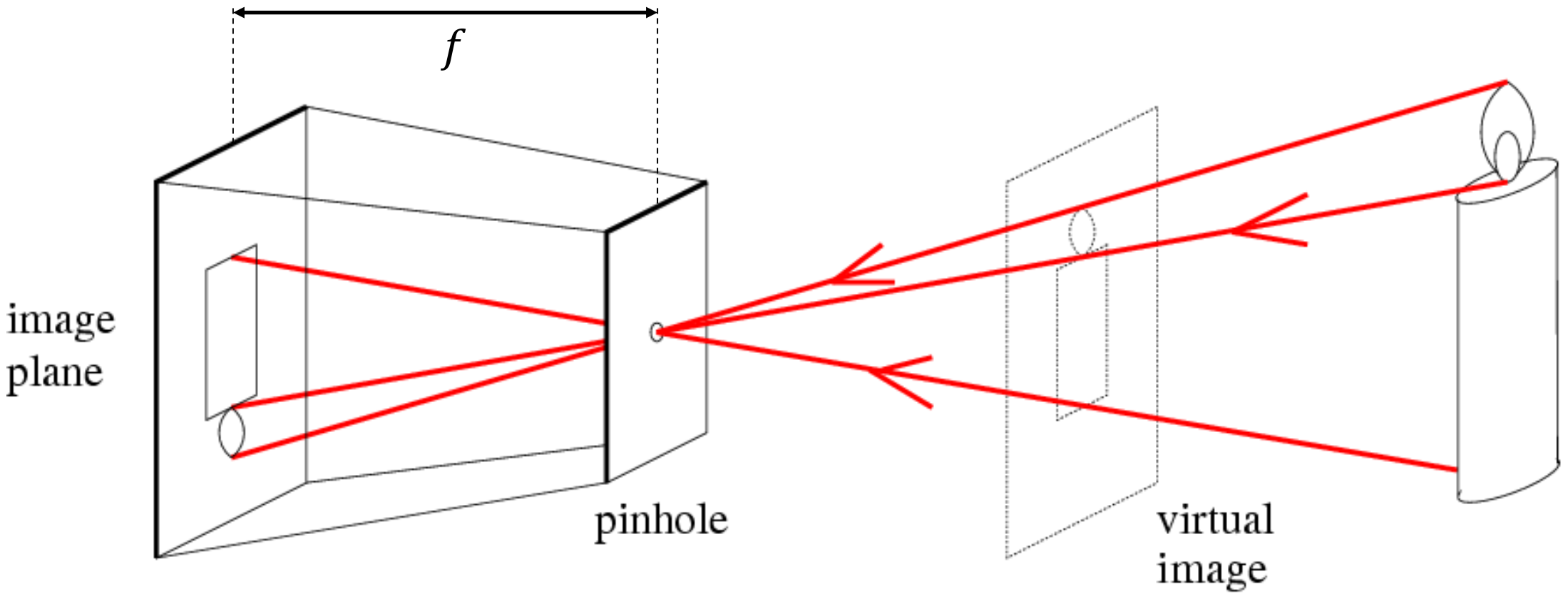


Illustration of "portable" camera obscura in Kircher's Ars Magna Lucis Et Umbrae (1645). Figure from wiki

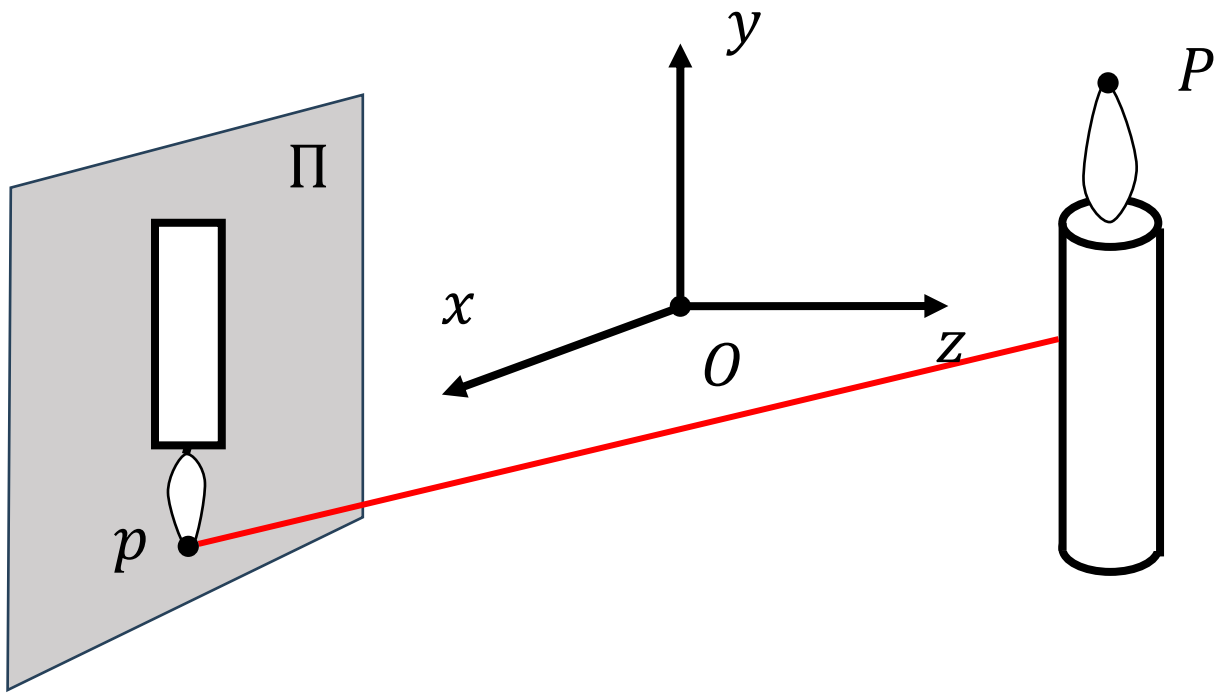
Pinhole Camera Model



Focal length f is the distance from the camera center
Pinhole / camera center / center of projection / aperture

Credit to David Forsyth

Modeling Projection



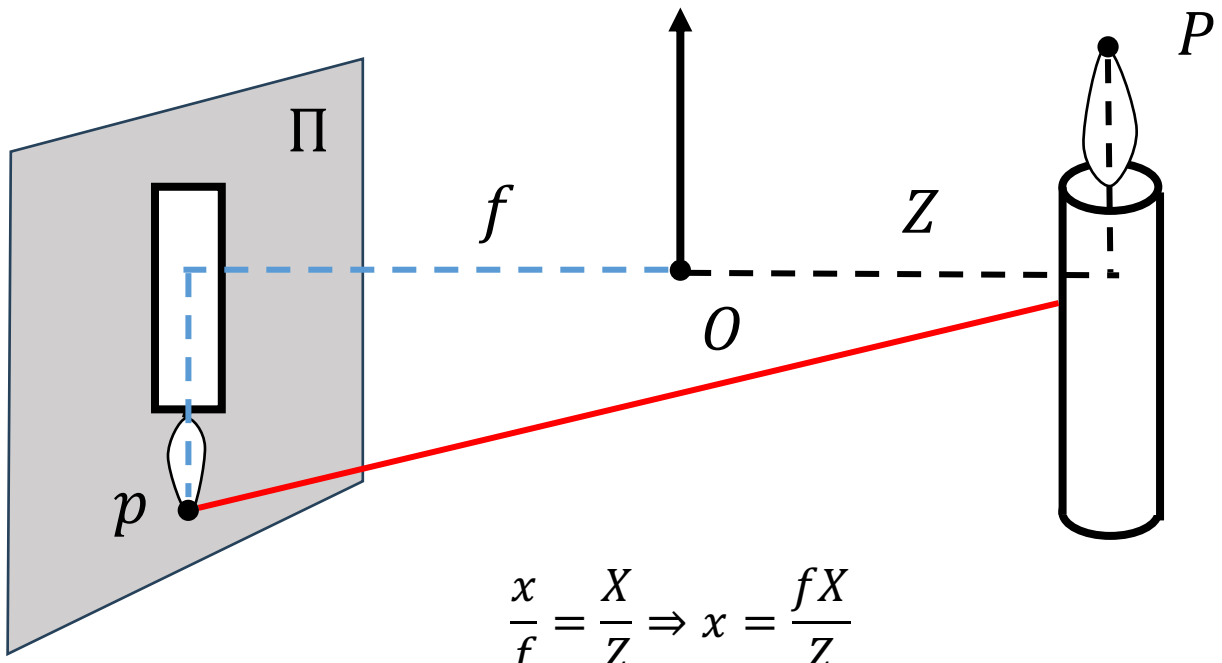
Terminologies

- optical/camera center O
- 3D point $P = (X, Y, Z)$
- image point $p = (x, y)$
- image plane Π
- optical axis (z-axis)
- focal length f

The coordinate system:

- The origin coincides with the pinhole
- The z-axis aligns with the optical axis

Perspective Projection Equation



$$\frac{x}{f} = \frac{X}{Z} \Rightarrow x = \frac{fX}{Z}$$

$$\frac{y}{f} = \frac{Y}{Z} \Rightarrow y = \frac{fY}{Z}$$

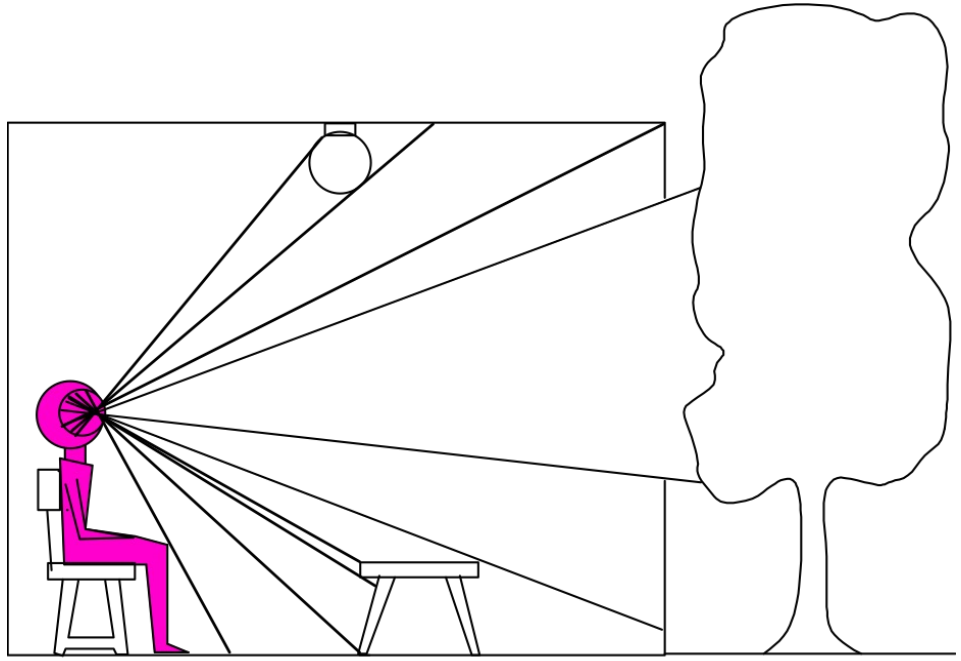
Terminologies

- optical/camera center O
- 3D point $P = (X, Y, Z)$
- image point $p = (x, y)$
- image plane Π
- optical axis (z-axis)
- focal length f

Projection equations derived by similar triangles: $(X, Y, Z) \rightarrow (f\frac{X}{Z}, f\frac{Y}{Z})$

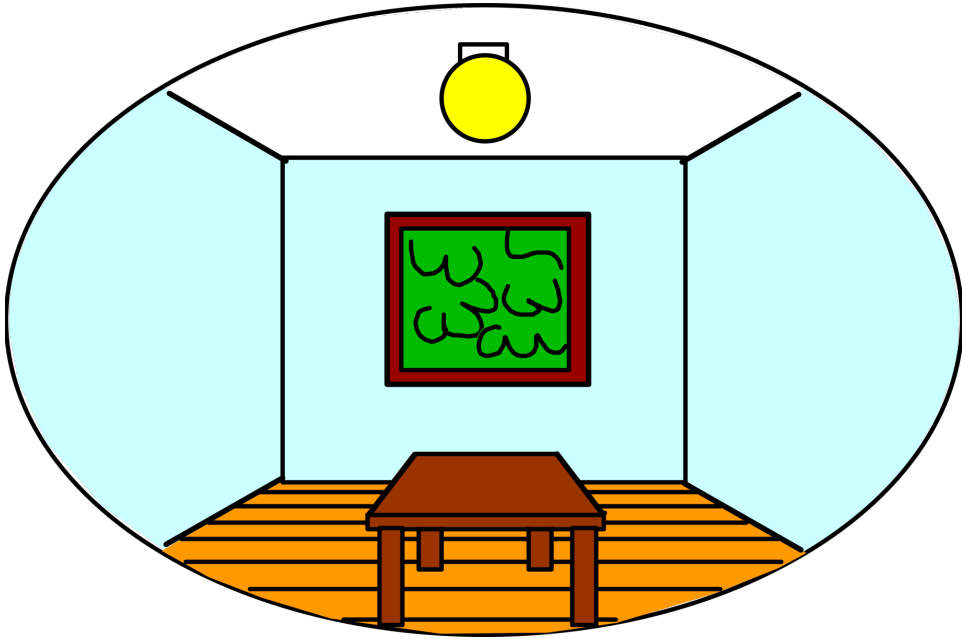
Projection as 3D to 2D (Dimensionality Reduction)

3D world



Point of observation

2D image



Figures © Stephen E. Palmer, 2002

Projection Can be Tricky



Meme: Why is the pigeon so large?

Figure from article

Projection Can be Tricky



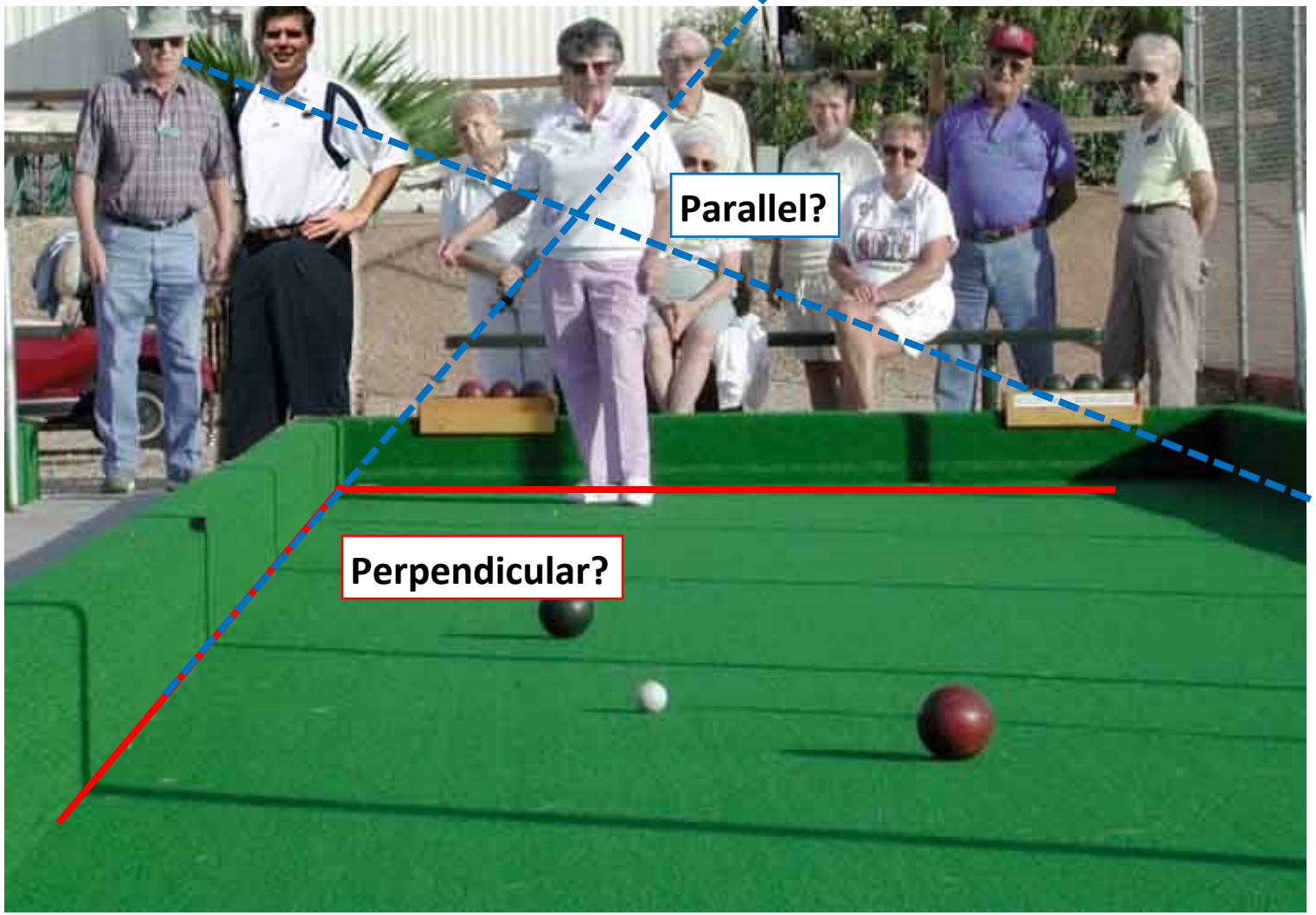
Is the globe actually spherical?

Projection Can be Tricky



Making of 3D sidewalk art: Youtube

What Else is not Preserved in Projective Geometry?



Angle is not preserved

What is Preserved?



Straight lines are still straight

Summary of Projective Geometry

- **Preserved**
  - points map to points
  - (most) 3D lines map to image lines
  - incidence and collinearity are preserved

- **Not preserved**
  - absolute length
  - angle
  - parallelism
  - metric depth
  - absolute size

More about projective geometry in later lectures

Case Study: Fronto-Parallel Planes

- What happens to the projection of a pattern on a plane parallel to the image plane?
  - All points on that plane are at a fixed depth z

- $(X, Y, Z) \rightarrow \left(f\ \frac{X}{Z}, f\ \frac{Y}{Z}\right) \rightarrow (X\ \frac{f}{Z}, Y\ \frac{f}{Z})$

- The pattern gets scaled by a factor of $\frac{f}{Z}$, but angles and ratios of lengths/areas are preserved

Camera with Lens

Part II: Why do real cameras need lenses?



Home-Made Pinhole Camera

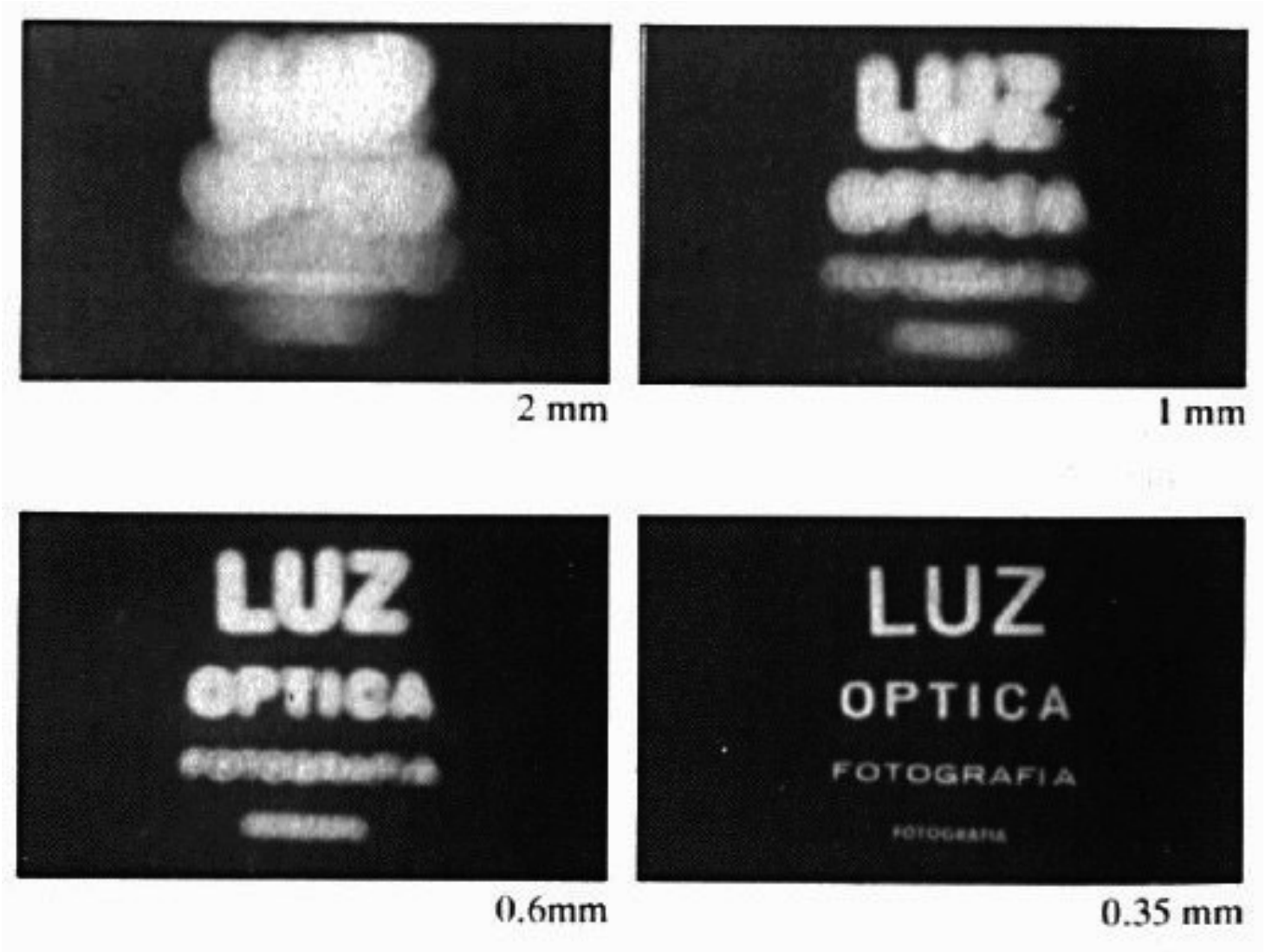


Figure from https://www.pauldebevec.com/Pinhole/

Why blurry?

In reality, the pinhole will have a finite size

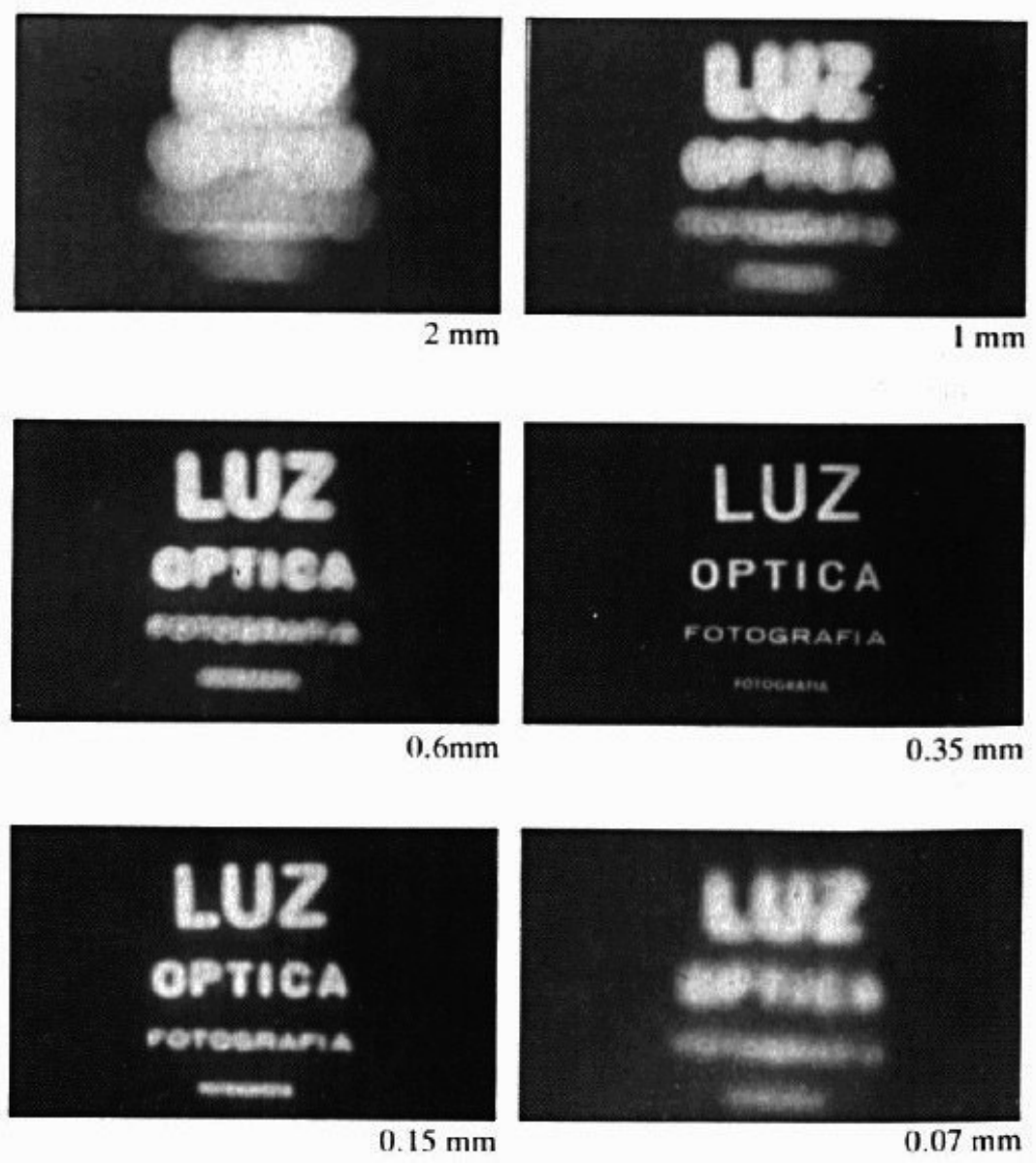
Shrinking the Aperture Produces Sharper Images



Why not make the aperture as small as possible?

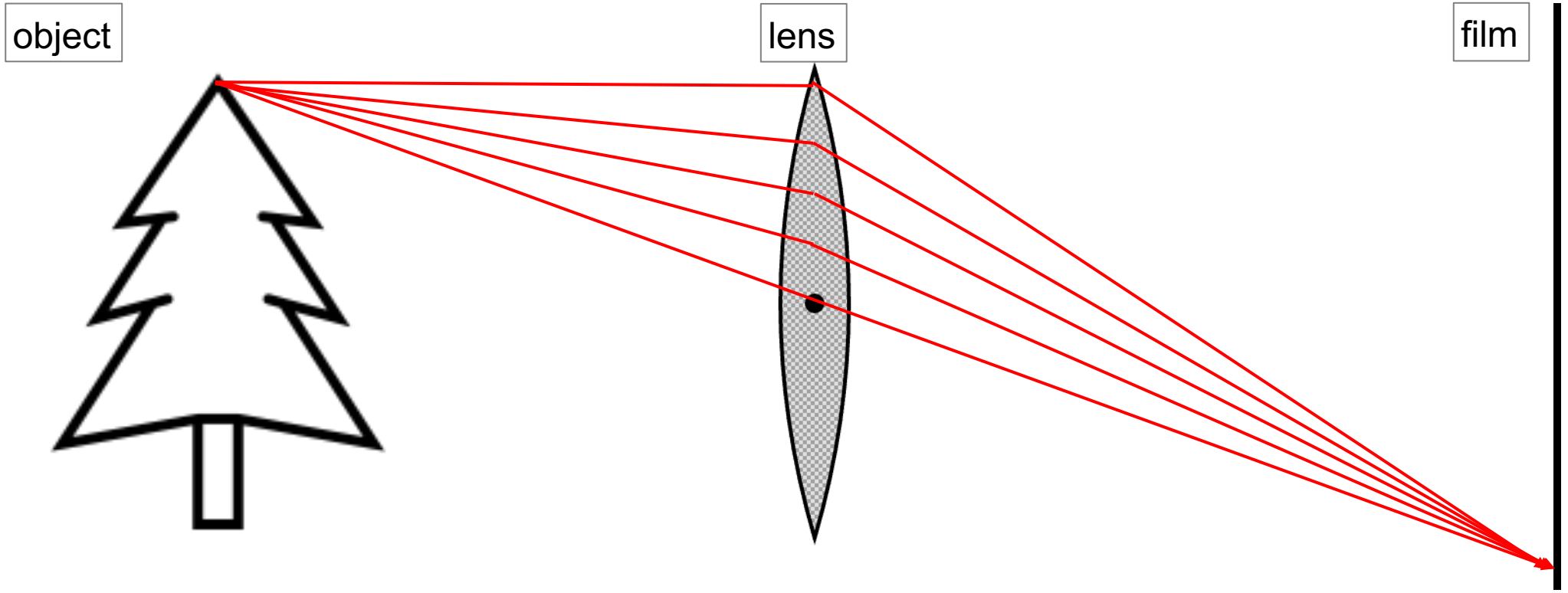
Credit to Steve Seitz

Shrinking the Aperture Reduces Brightness



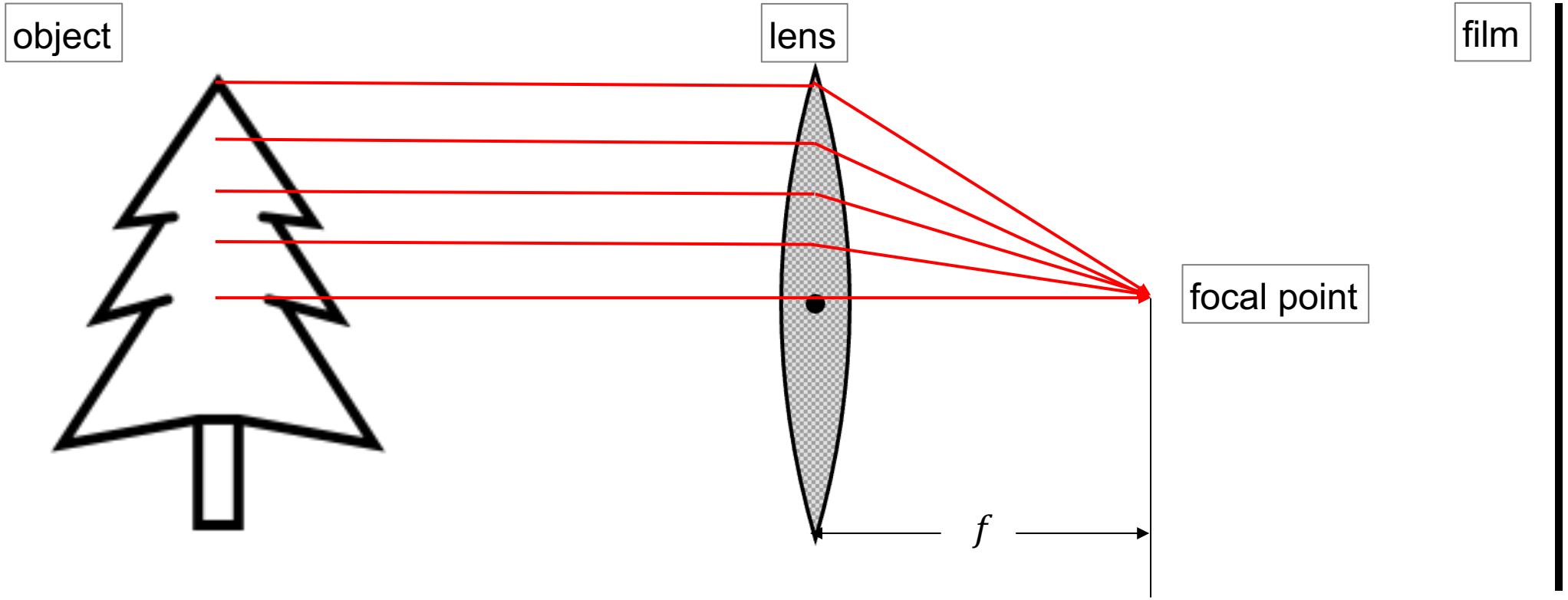
Shrinking the pinhole produces sharper images but reduces the amount of light reaching the image plane, and may introduce *diffraction* effects.

Solution: Adding a Lens



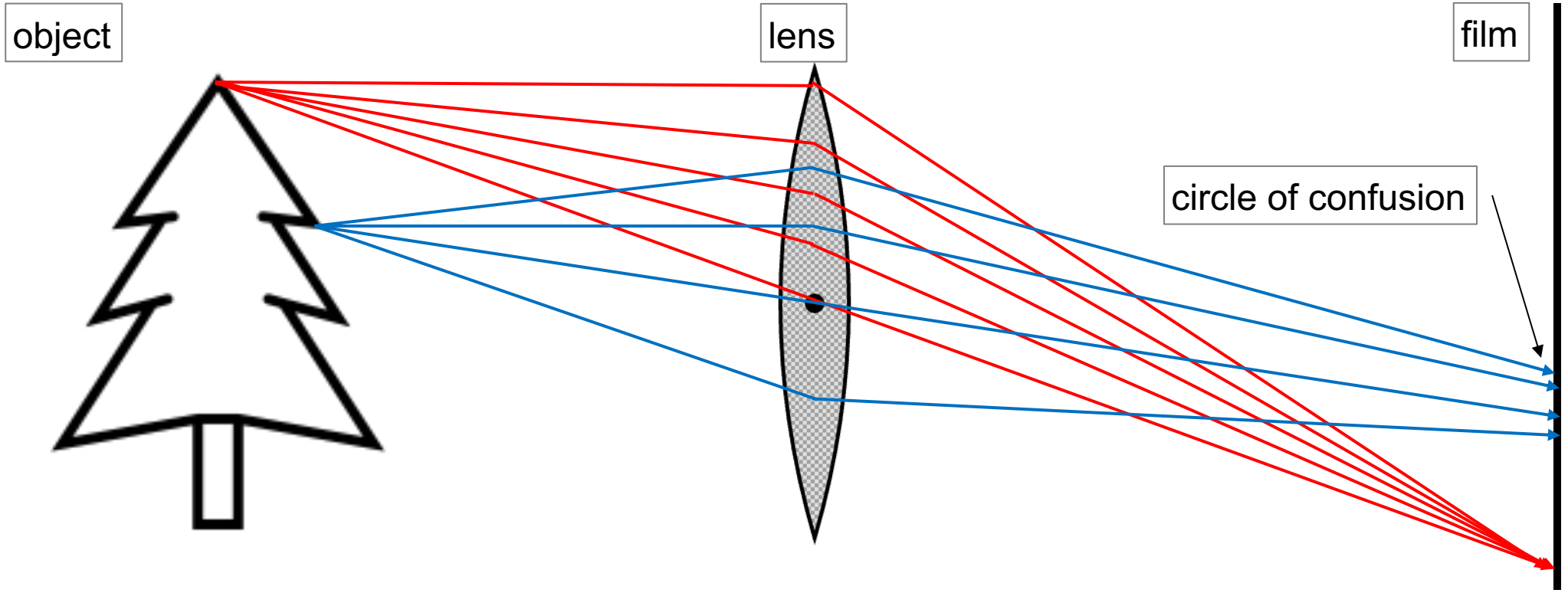
In the ideal thin-lens model, a ray through the optical center is modeled as undeviated
A lens admits many rays from one scene point and brings them together at one image point

Focal Length and Focal Point



Rays parallel to the optical axis converge at the focal point

In Focus

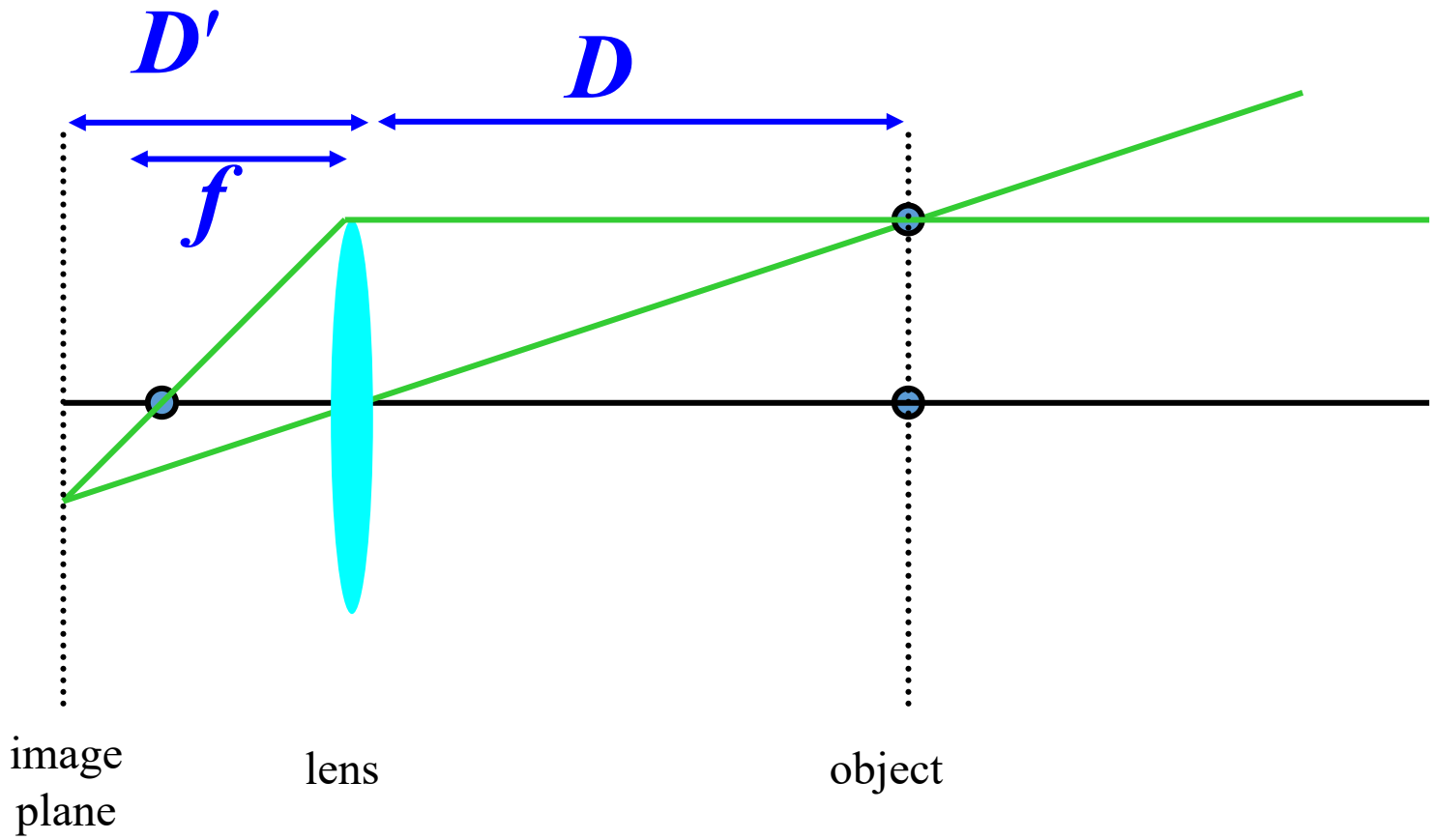


For a fixed sensor position, one object distance is exactly in focus.
Points at other depths form nonzero blur circles on the sensor.

Circle of confusion: the sensor-plane footprint produced by an out-of-focus scene point.

Thin Lens Formula

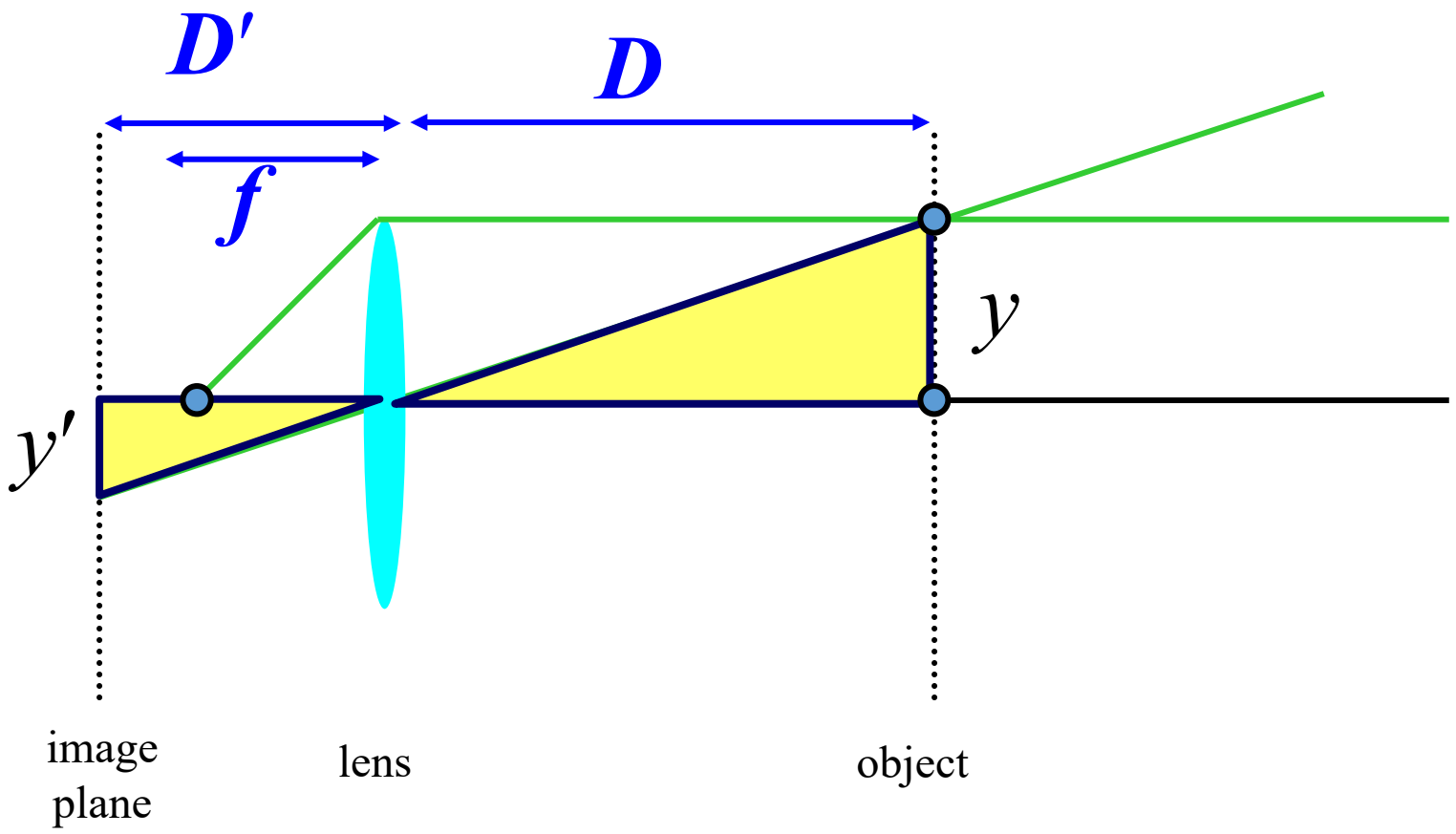
What is the relation between the focal length (f),
the distance of the object from the optical center (D),
and the distance at which the object will be in focus (D')?



Thin Lens Formula

Similar triangles

$$y'/y = D'/D$$

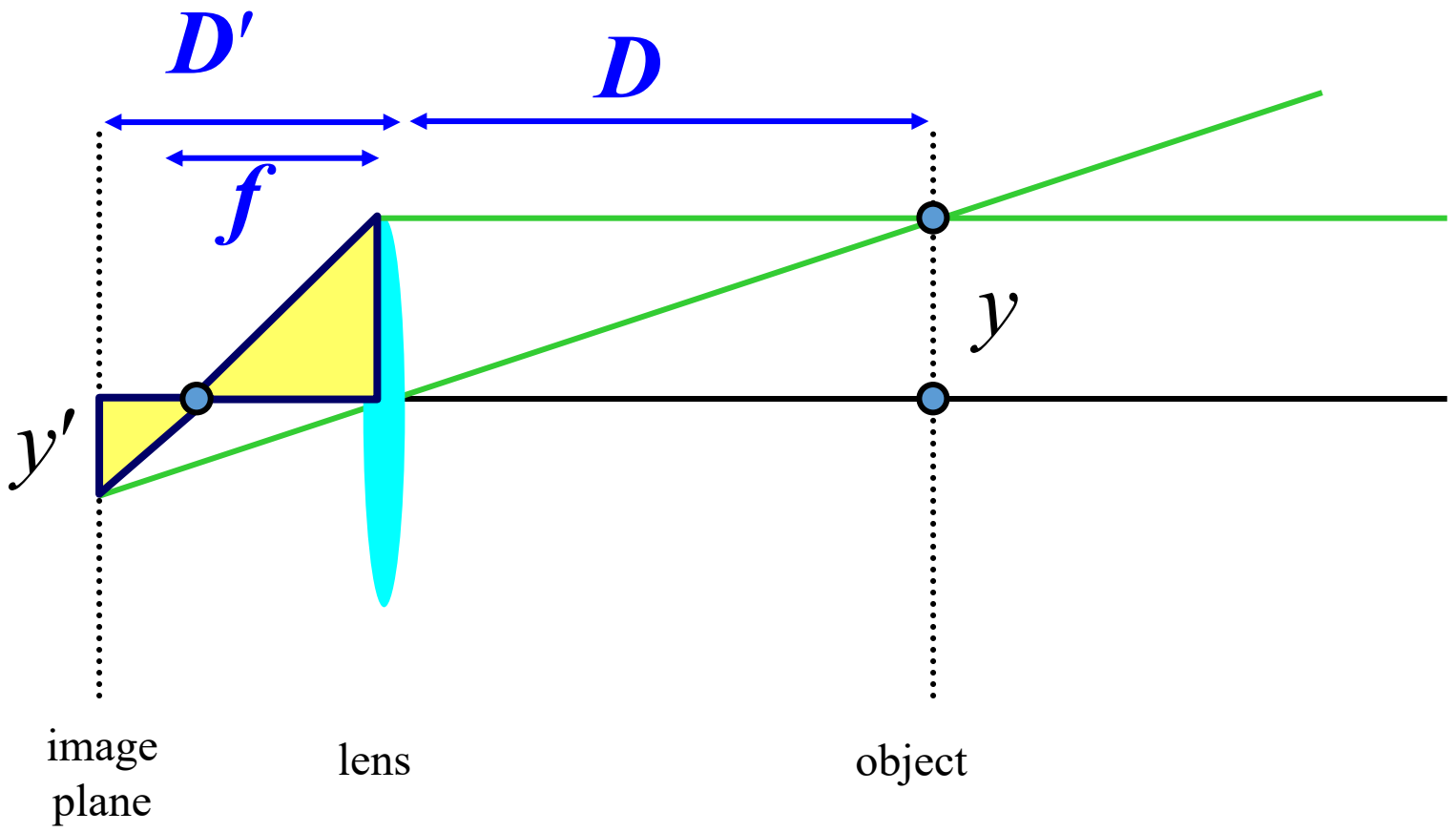


Thin Lens Formula

Similar triangles

$$y'/y = D'/D$$

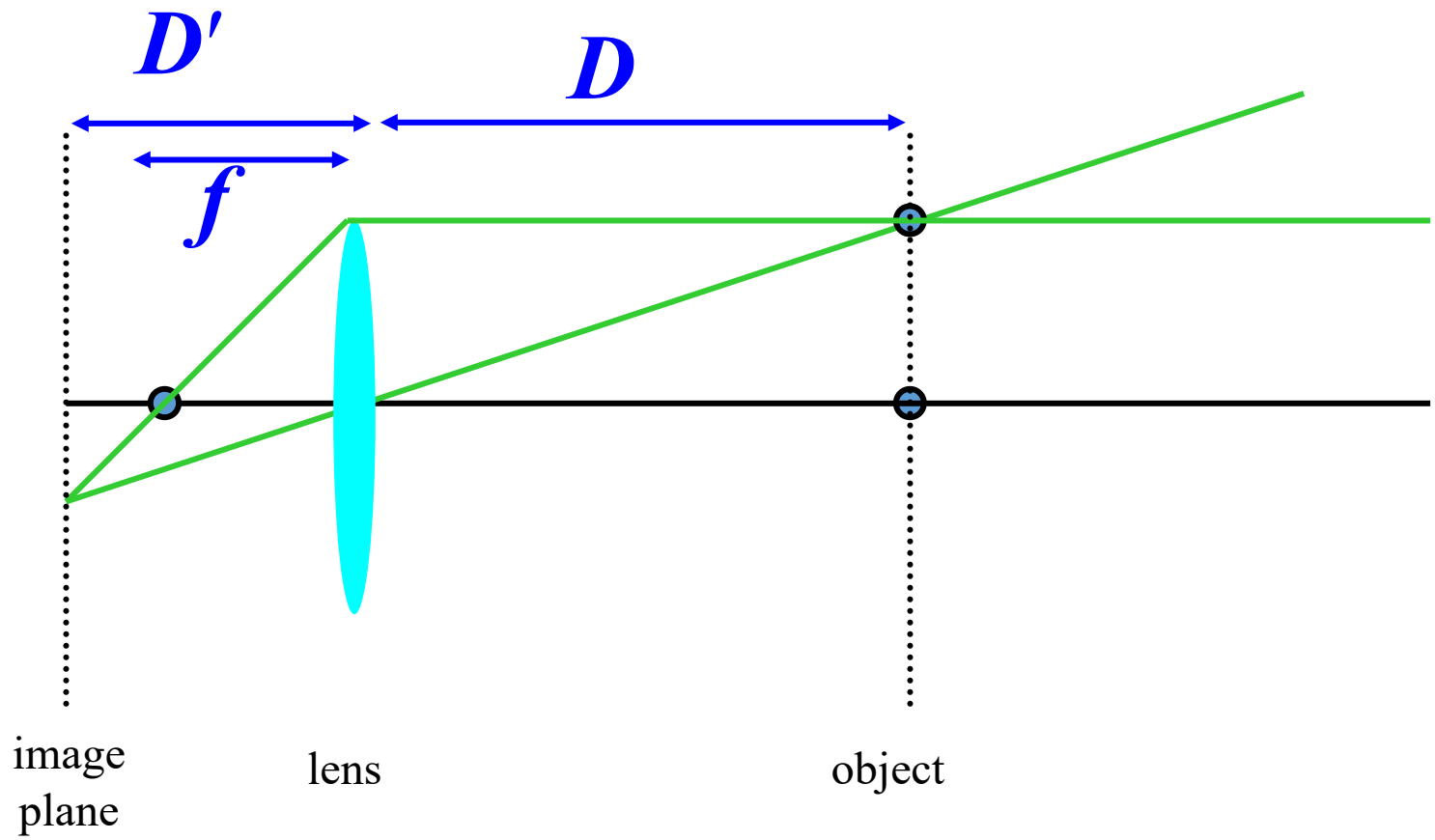
$$y'/y = (D' - f)/f$$



Thin Lens Formula

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

For an object at distance D, a sharp image forms at sensor distance D' satisfying the thin-lens equation.



Approximation of Projection Equations

- When does the projection equations still hold given the thin-lens formula?
  - $D \to \infty \Rightarrow D' \to f$
  - The camera is focused at infinity

- Next, we will see how the effects of depth, focal length and aperture

Varying Focus



(a)



(b)



(c)

Credit to A. Efros

Depth of Field (DoF)

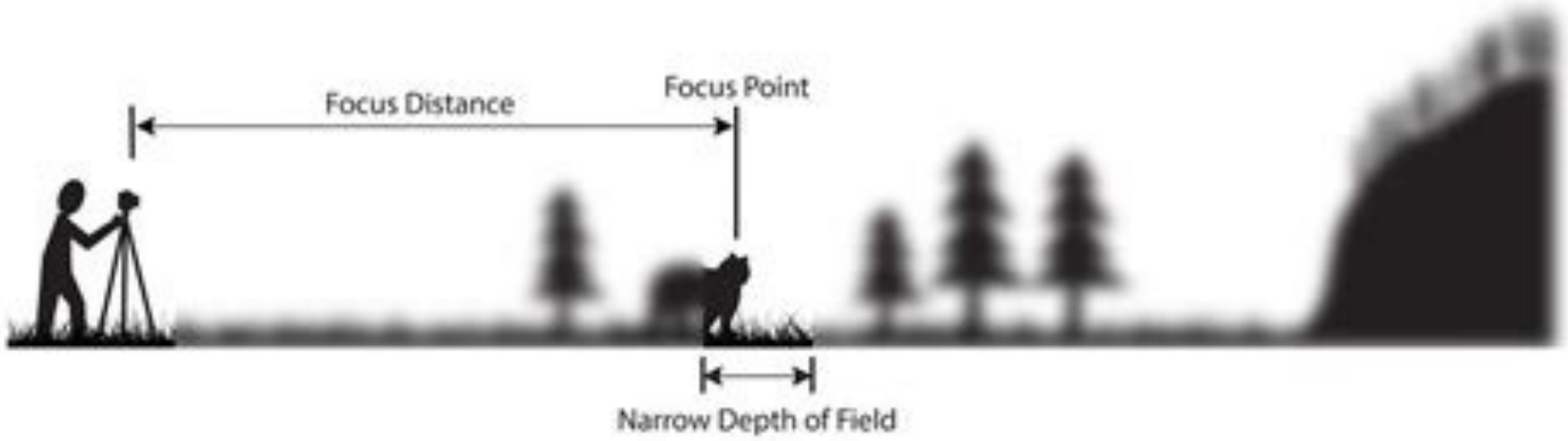


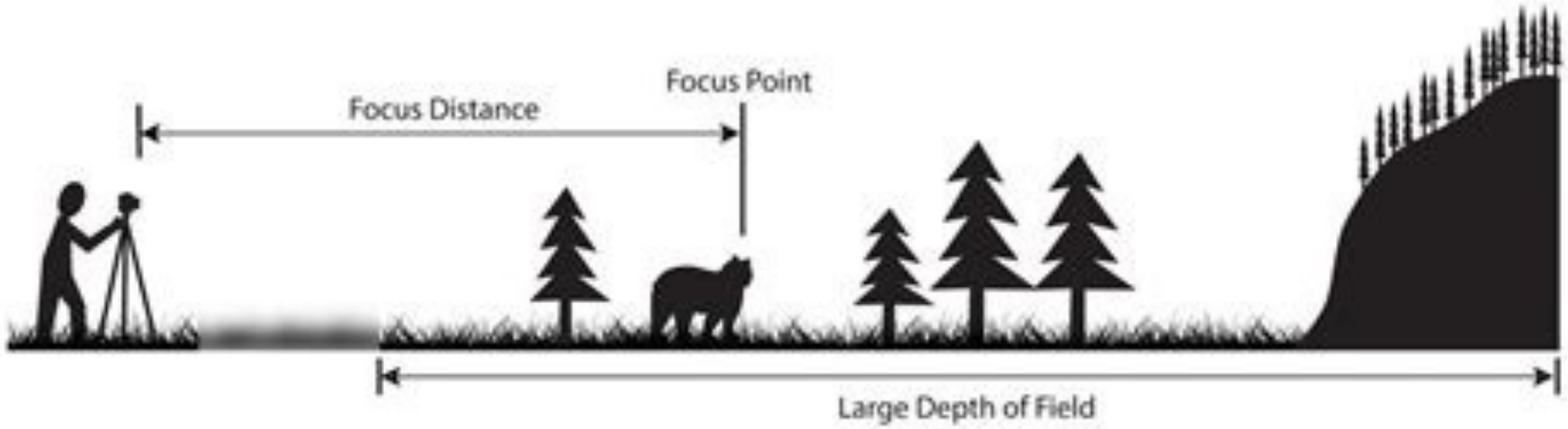
Depth of field is the range of object distances whose blur remains below an accepted circle-of-confusion threshold.

http://www.cambridgeincolour.com/tutorials/depth-of-field.htm

Credit to A. Efros

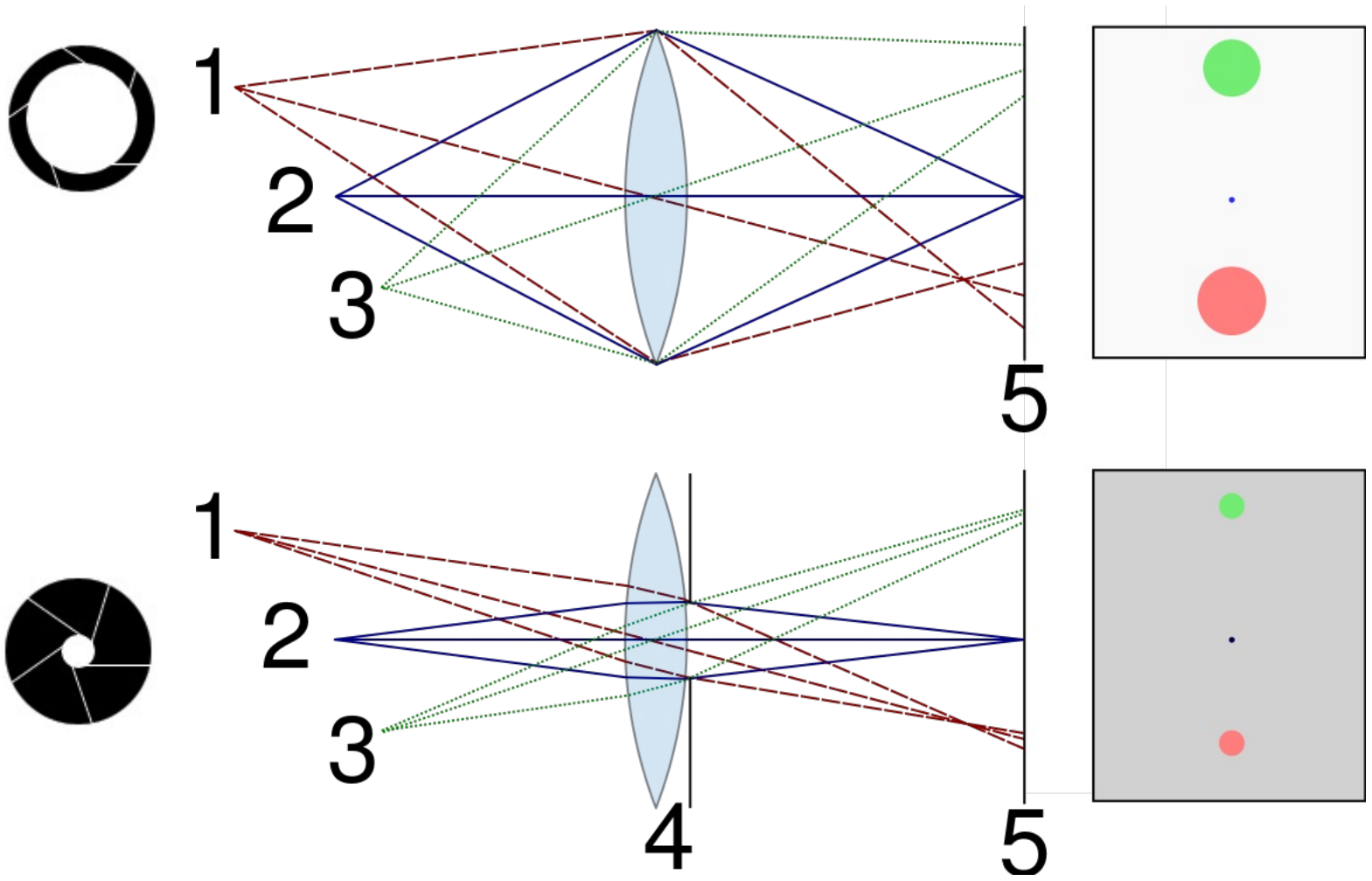
Different Depth of Field





How to control the depth of field?

Effect of Aperture on DoF



A smaller aperture increases the range in which the object is approximately in focus

But small aperture reduces amount of light – need to increase exposure

From wiki

Effect of Aperture on DoF



Aperture = *f*/1.4. DOF = 0.8 cm | Aperture = *f*/4.0. DOF = 2.2 cm | Aperture = *f*/22. DOF = 12.4 cm

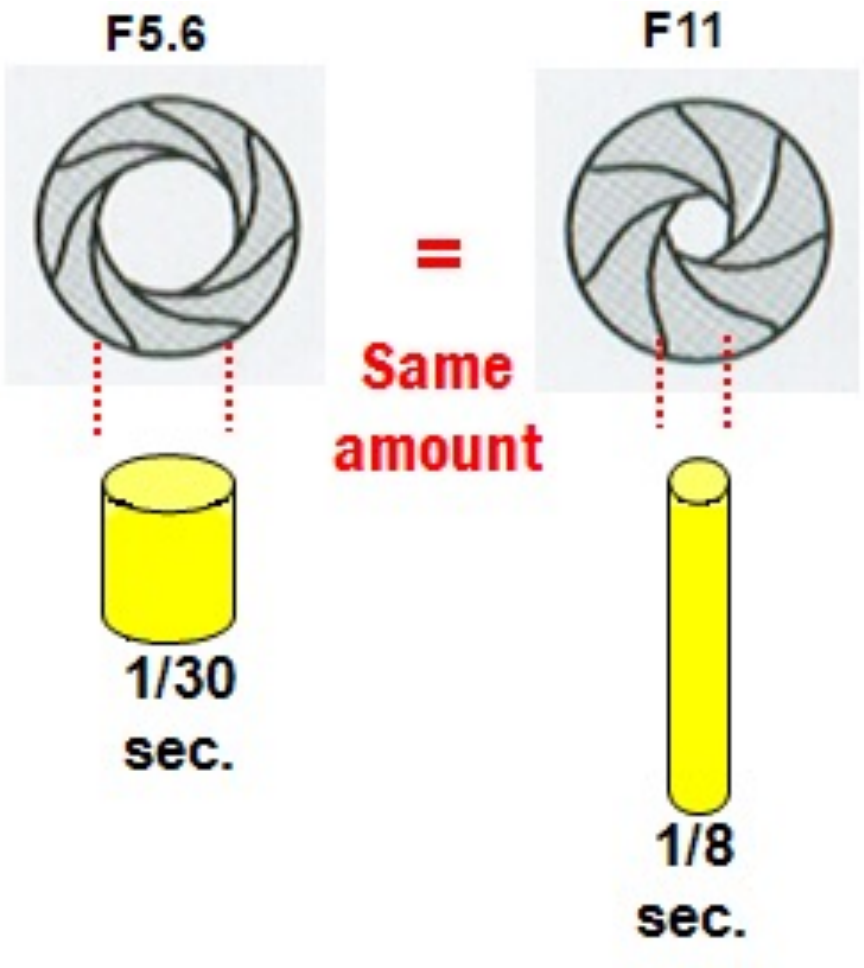
Depth of field for different values of aperture using 50 mm objective lens and full-frame DSLR camera. Focus point is on the first blocks column.

f-number: the ratio of lens focal length to aperture diameter

$$\frac{f}{\#} = N = \frac{f}{d}$$

From wiki

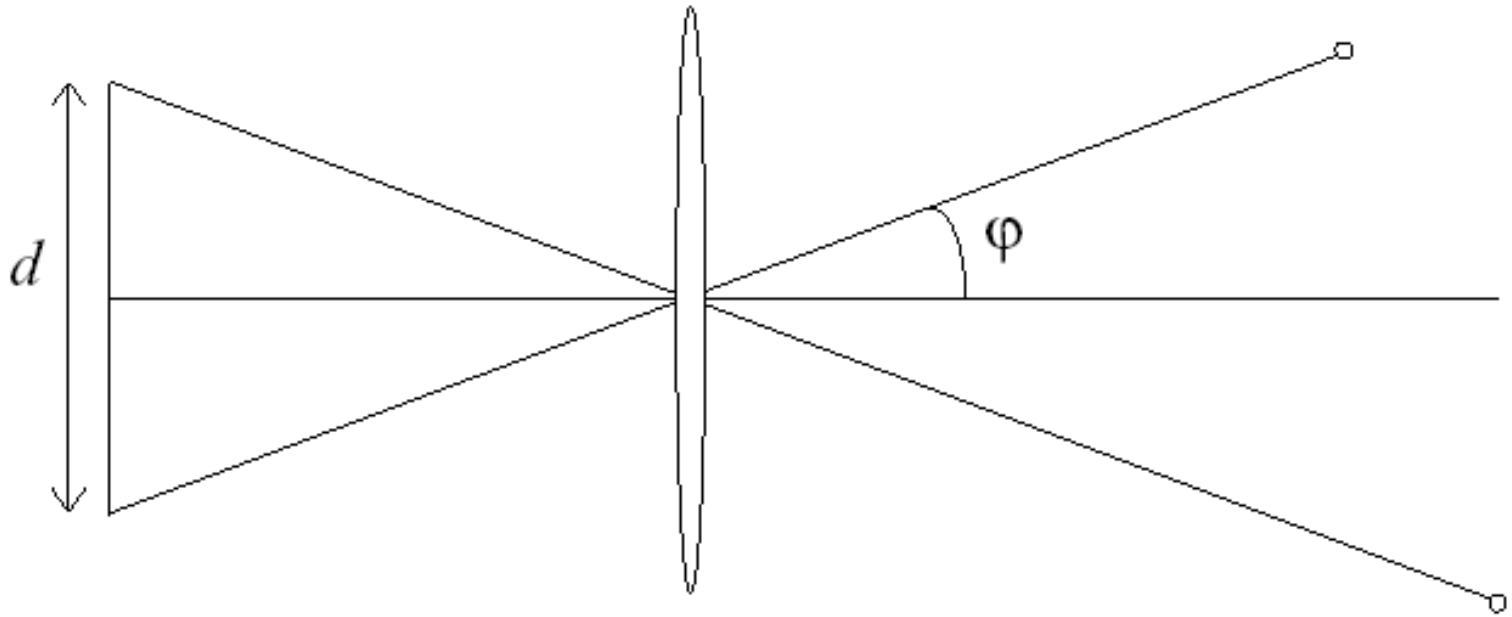
Exposure





Credit to A. Efros

Field of View (FOV)

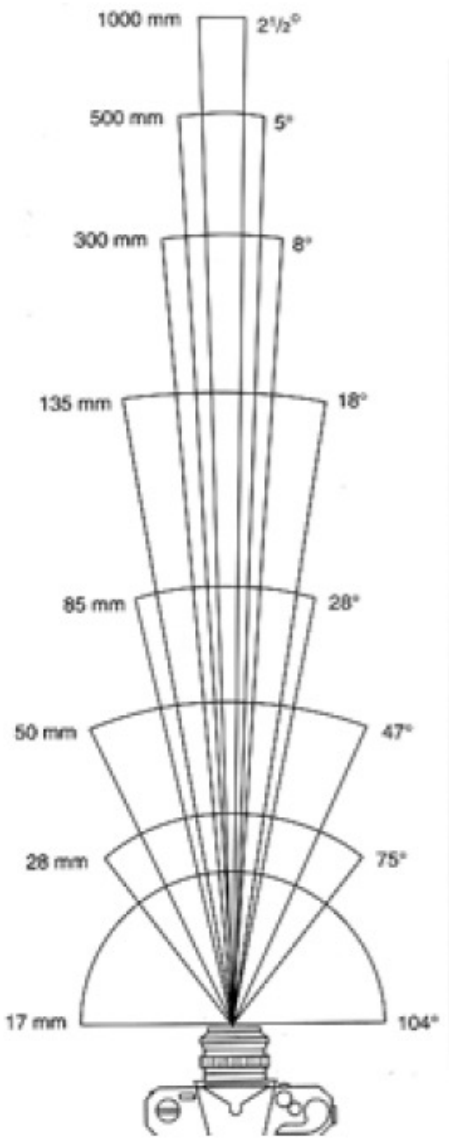


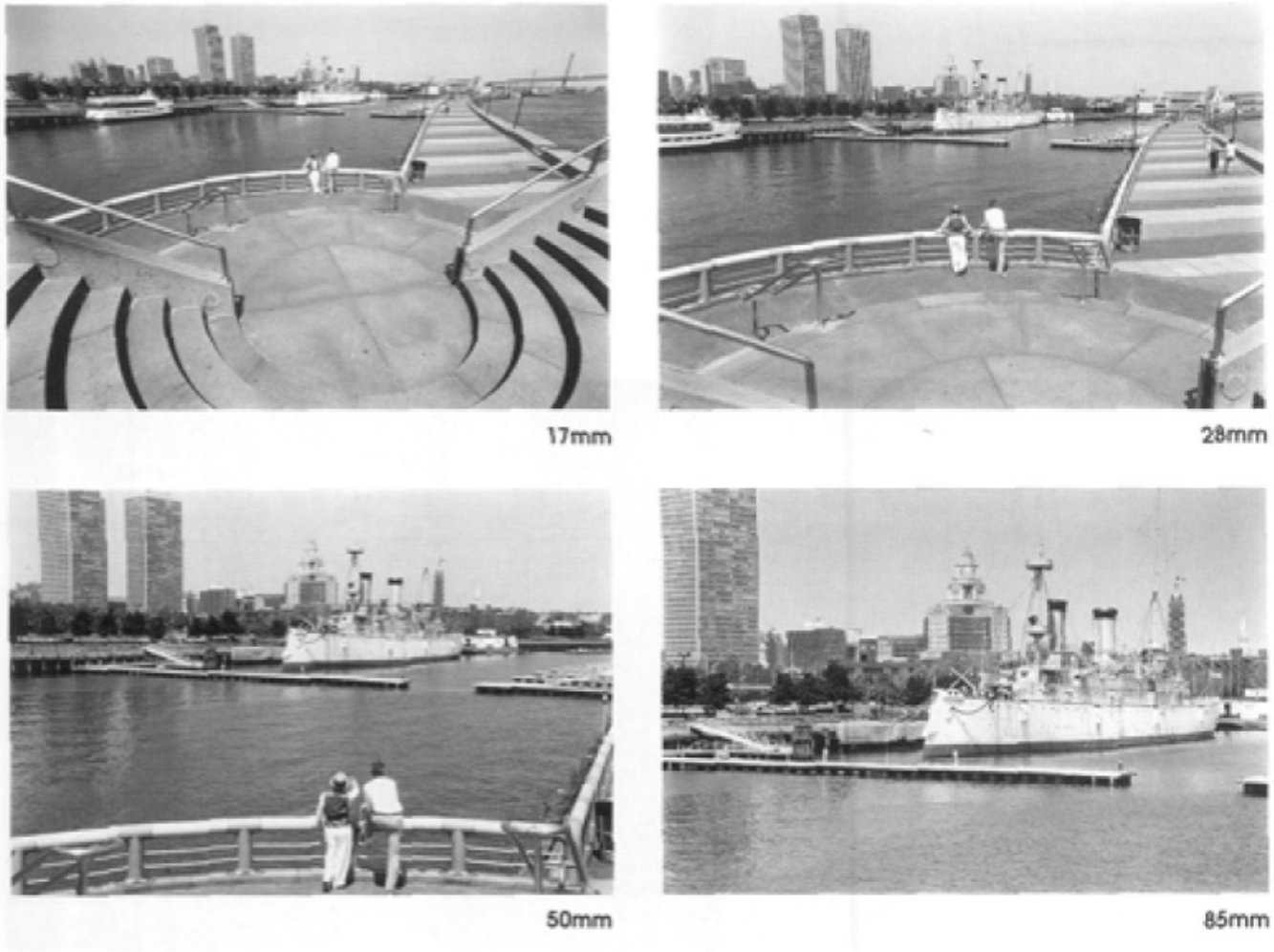
FOV (2φ) depends on focal length and size of the camera sensor.

$$\varphi = \tan^{-1}(\frac{d}{2f})$$

Larger focal length gives smaller FOV for a fixed sensor size

Changing Field of View (Zoom)





From London and Upton

Credit to A. Efros

Increasing Focal Length



Sigma 200-500mm F2.8 EX DG lens

What does 1600mm lens look like?

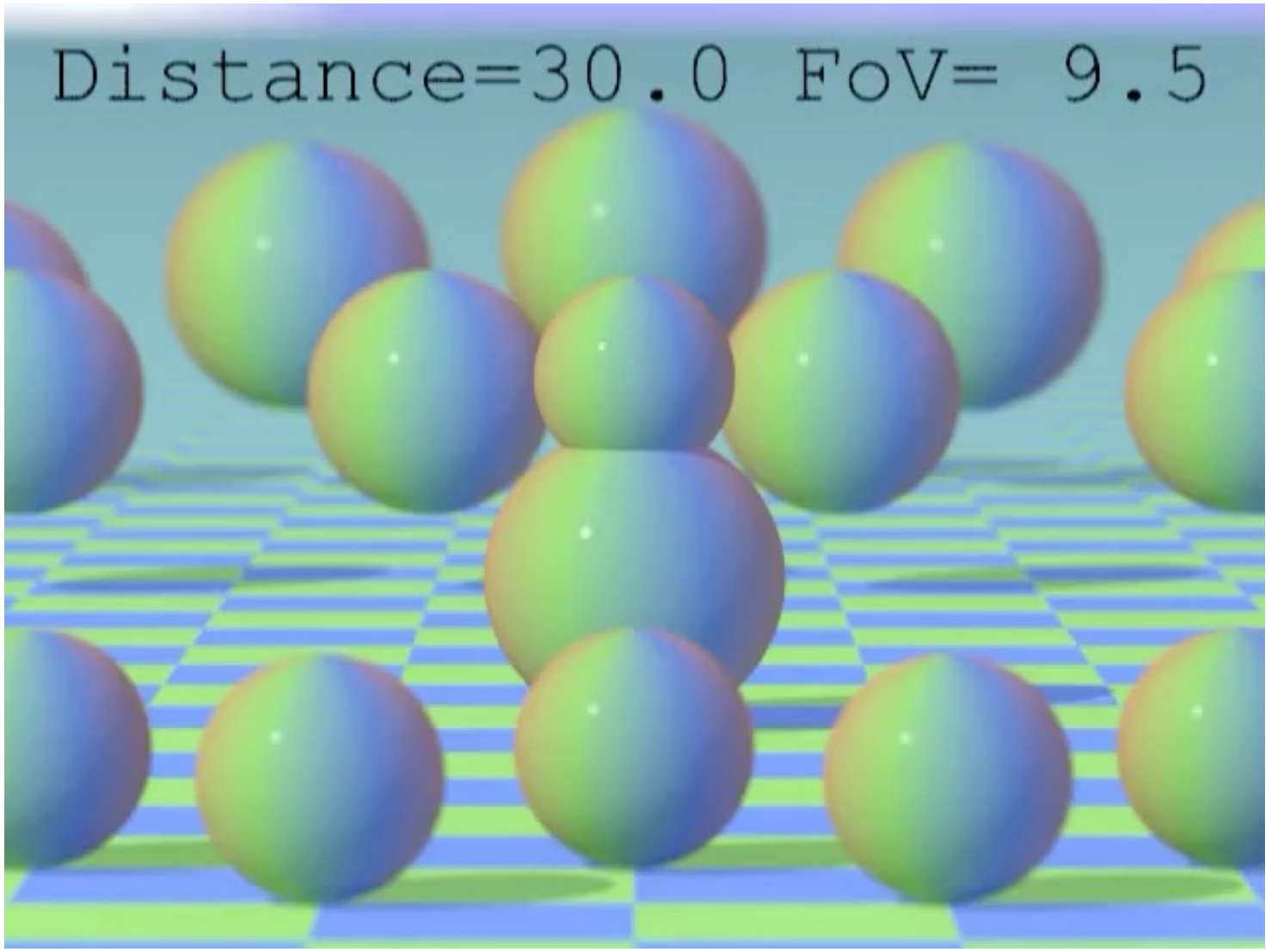
http://www.digitalpixels.net/varia/the-web/sigma-200-500mm-f28-ex-dg-lens-on-the-field/



http://dancarrphotography.com/blog/wp-content/uploads/2011/05/Canon_super_tele_comparison.jpg

Credit to A. Efros

Dolly Zoom



"subject image size" $\propto \frac{fH}{Z}$

Continuously adjusting the focal length while the camera moves away from (or towards) the subject

From wiki

Dolly Zoom – The Vertigo Shot

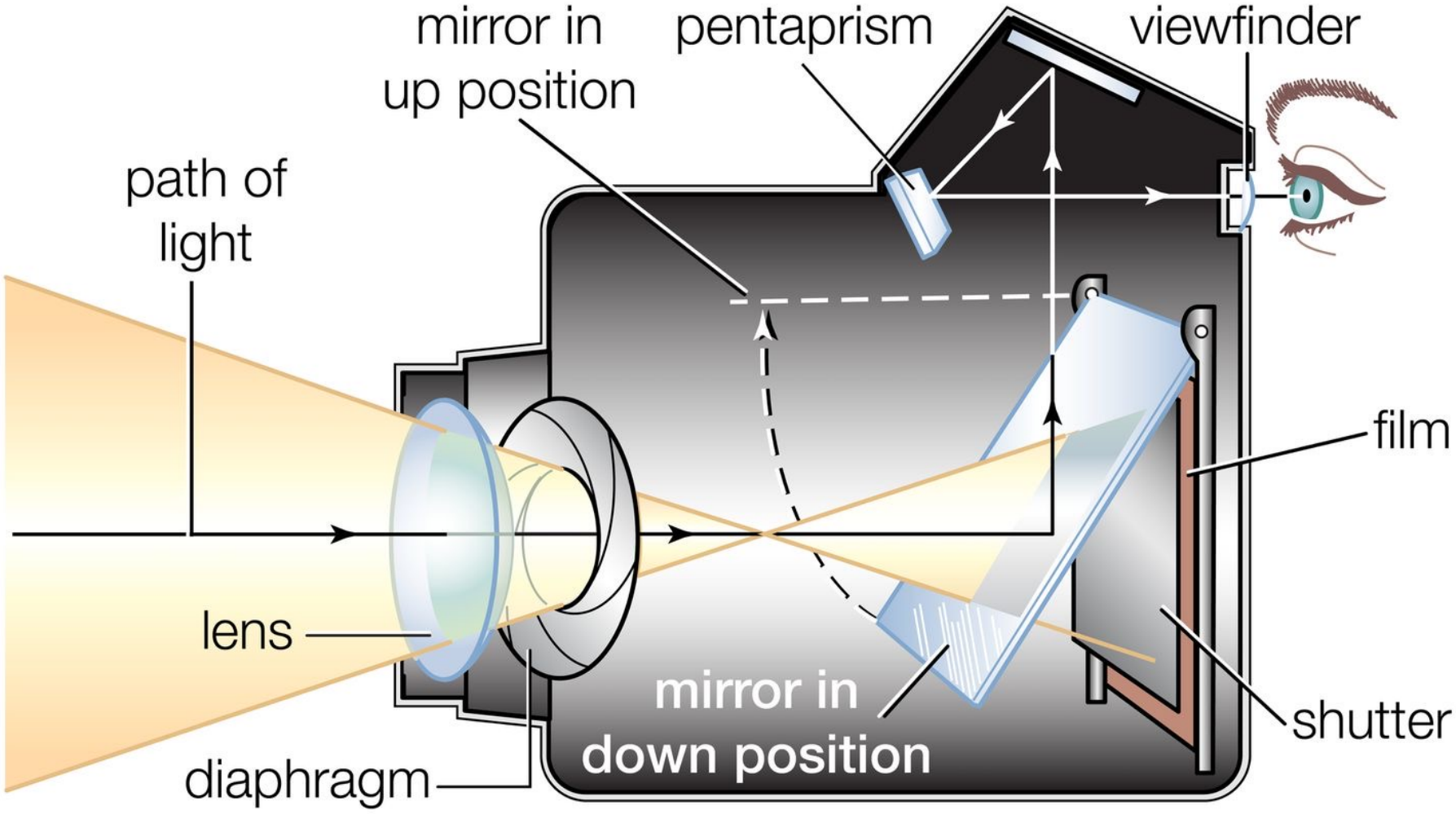


Dolly Zoom Effect (Vertigo 1958)

Lens Flaws

Part III: Where does the ideal camera model fail?

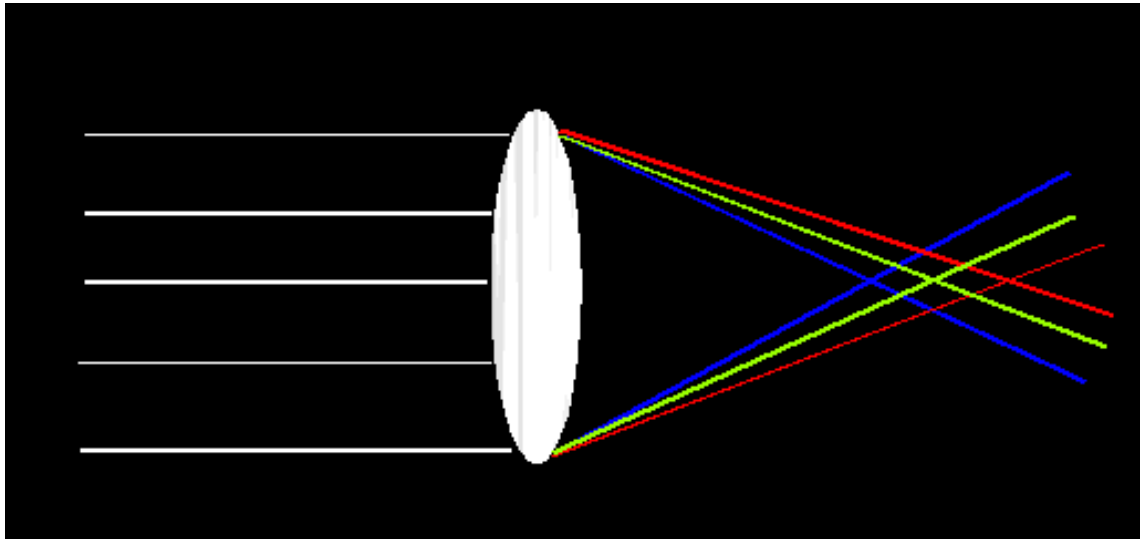
Building a Real Camera



scene→lens→aperture→shutter→sensor→digital values

https://www.thesprucecrafts.com/what-to-know-about-camera-body-2688630

Lens Flaws 1: Chromatic Aberration





Lens has different refractive indices for different wavelengths. Thus, light of different colors focus at slightly different distances

Lens Flaw 2: Vignetting

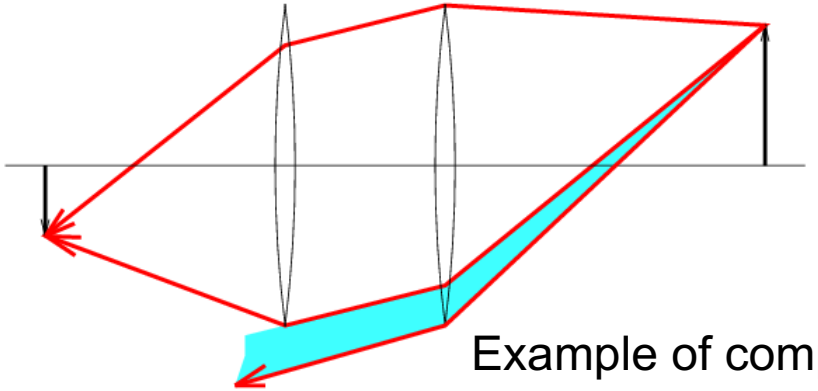
Vignetting is a decrease in recorded brightness toward the image boundaries





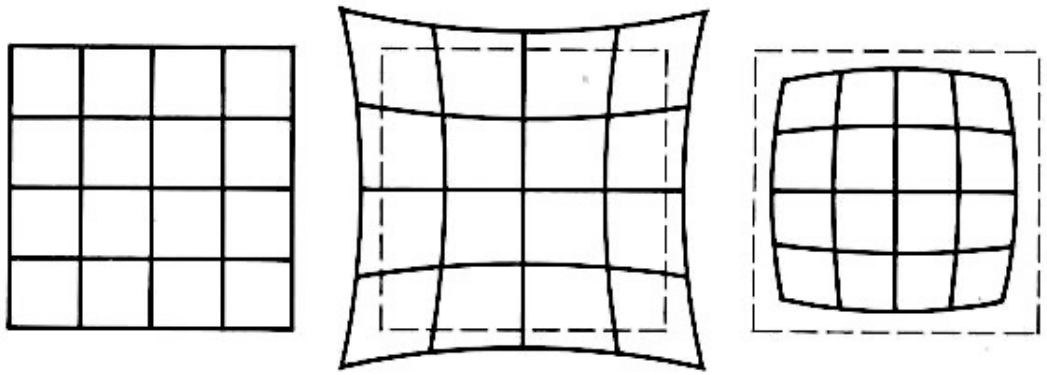
Two main types:

- natural vignetting: due to the foreshortening in the object surface, projected pixel, and lens aperture
- mechanical vignetting: caused by the internal occlusion of rays near the periphery of lens elements in a compound lens



Example of compound lens

Lens Flaws 3: Radial Distortion



No distortion Pin cushion Barrel

Radial distortion is a systematic deviation from ideal perspective projection caused by lens design



Distortion displacement grows with distance from the distortion center

Lines may not be mapped to lines in this case

Sometimes, Lens Distortions are Expected



Fisheye camera

Figure 2.13c in CVAA

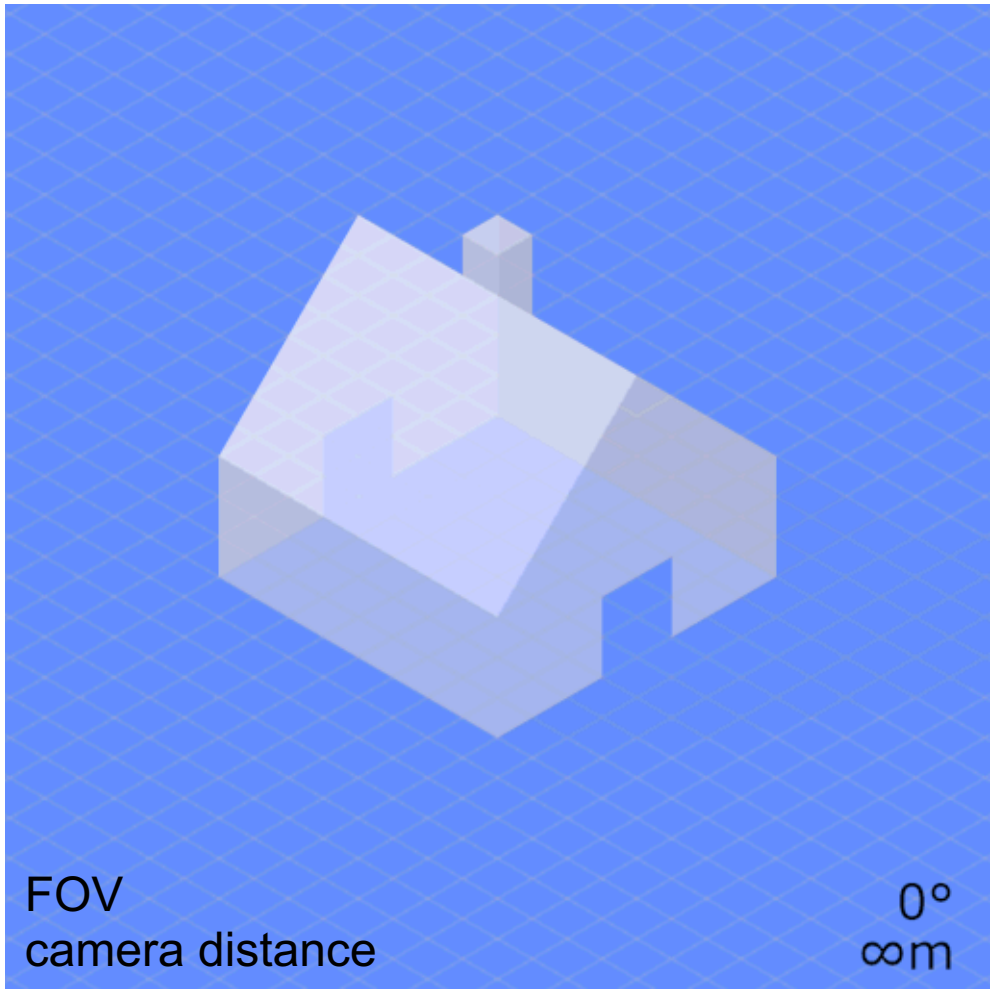
Perspective Distortion: Model or Lens Defect?

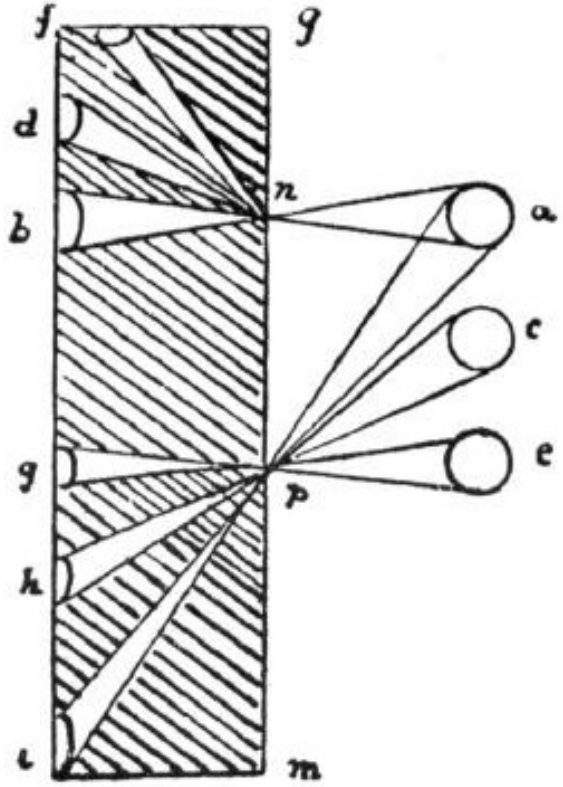


How to relieve the perspective effects on the nose?
How to adjust the focal length and the camera-subject distance?

Credit to A. Efros

Perspective Distortion





Phenomenon pointed out by Da Vinci

From wiki

Perspective Distortion Comes from Viewpoint

What changes perspective? Focal length or camera position?

Consider two 3D points (X, Y, Z) and $(X + \Delta X, Y, Z + \Delta Z)$

Their projected x coordinates:

$$x = f\frac{X}{Z} \qquad x + \Delta x = f\frac{X + \Delta X}{Z + \Delta Z}$$

Their projected distance (x-axis):

$$\Delta x = f\frac{X + \Delta X}{Z + \Delta Z} - f\frac{X}{Z} = f\frac{Z\Delta X - X\Delta Z}{(Z + \Delta Z)Z}$$

Q1: Given the same object position and focal length, what if we change the camera position (Z)?

Q2: Given the same camera position and focal length, what if we change the object position (X)?

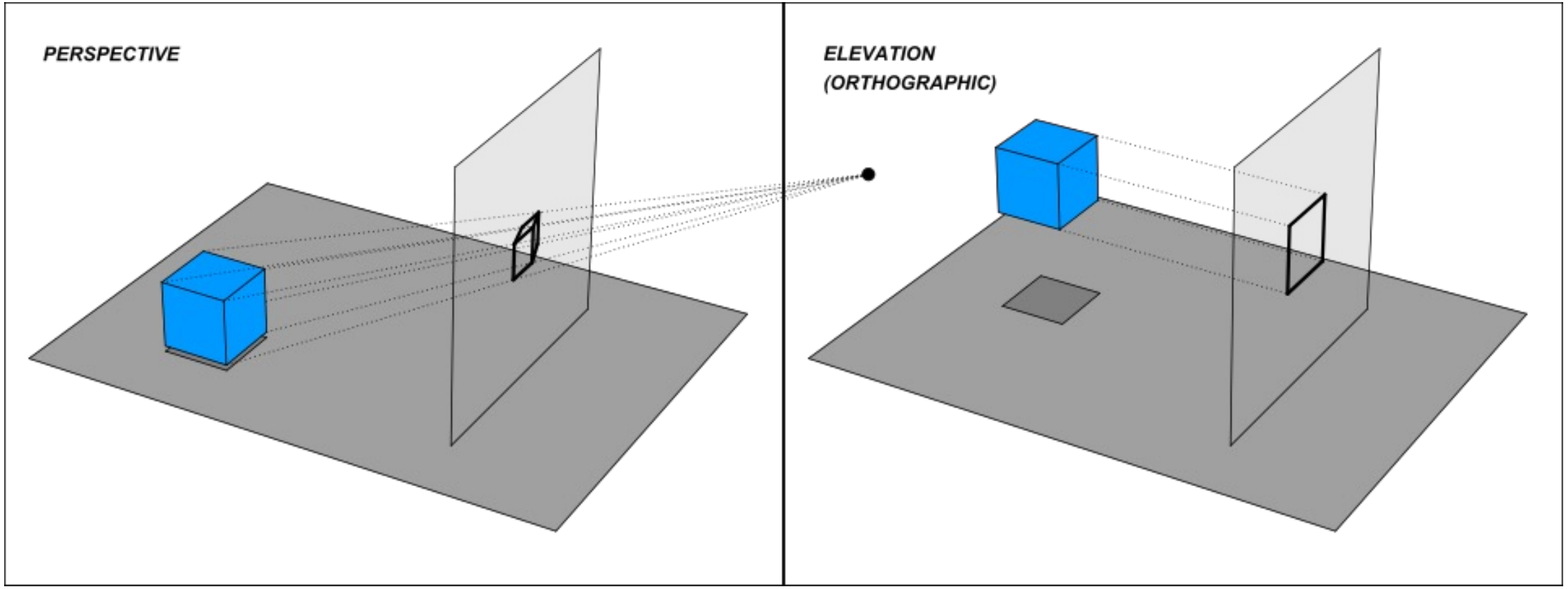
Weak Perspective or Scaled Orthography

When a scene's relief is small relative to its average distance from the camera, the magnification can be taken to be constant (reduced to the *fronto-parallel* case).

If the relative depth difference of an object is much smaller than the average camera-object distance: $\Delta Z \ll Z_0$

$$x = f\frac{X}{Z_0 + \Delta Z} \approx f\frac{X}{Z_0} = \frac{f}{Z_0}X$$

Orthographic (Parallel) Projection



$$(X, Y, Z) \rightarrow \left(f\,\frac{X}{Z}, f\,\frac{Y}{Z}\right)$$

$$(X, Y, Z) \rightarrow (X, Y)$$

Orthographic projection preserves size across depth, because image coordinates do not depend on Z

Figure from wiki

Summary

- Concepts
  - Pinhole camera
  - Focal length, aperture
  - Depth of field and field of view
  - Lens flaws
  - Perspective distortion
- Computation
  - Perspective projection
  - Thin-lens formula

- What is the role of the pinhole?
- What is the role of the lens?
- What is dolly effect?
- What affect perspective distortion?

Reading List

- **Szeliski, Ch. 2 — Image Formation**
  - Ch. 2.1.4 3D to 2D Projections
  - Ch. 2.1.5 Lens Distortions
  - Ch. 2.2.3 Optics
- **Forsyth & Ponce, Ch. 1 — Geometric Camera Models**
  - Ch. 1.1.1–1.1.3 Pinhole / Weak Perspective / Lenses
  - Ch. 1.2.2–1.2.5 Intrinsics / Extrinsics / Projection Matrices
- Stanford CS231A — Camera Models
- Berkeley CS180/280A — Course materials